Project Executable Files

Date	15 July 2026
Team ID	SWTID-2026-2749
Project Name	OptiCrop: Smart Agricultural Production Optimization engine
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	NA
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```
OptiCrop/
│
├── app.py
├── requirements.txt
├── README.md
├── .gitignore
│
├── analysis/
│   ├── dataset_analysis.py
│   ├── univariate_analysis.py
│   ├── bivariate_analysis.py
│   ├── multivariate_analysis.py
│   └── graphs/
│
├── dataset/
│   └── crop_recommendation.csv
│
├── models/
│   ├── logistic_regression_model.py
│   ├── decision_tree_model.py
│   ├── random_forest_model.py
│   ├── knn_model.py
│   ├── kmeans_model.py
│   └── elbow_method.py
│
├── preprocessing/
│   ├── outlier_detection.py
│   ├── seasonal_analysis.py
│   └── train_test_split_data.py
│
├── saved_models/
│   ├── random_forest_model.pkl
│   ├── decision_tree_model.pkl
│   ├── logistic_regression_model.pkl
│   ├── knn_model.pkl
│   └── scaler.pkl
│
├── static/
│   ├── css/
│   ├── js/
│   ├── images/
│   └── crops/
│
├── templates/
│   ├── index.html
│   ├── about.html
│   ├── contact.html
│   └── prediction.html
│
└── venv/
```

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Step 3: Deployment / Access Details

Field	**Details**
Hosted / Deployed URL	Not Hosted (Runs Locally)
Login Credentials (Demo)	Not Required
Platform / Hosting Provider	Local Flask Development Server
Repository Link	https://github.com/shaikmubi224/OptiCrop.git
Demo Video Link	https://drive.google.com/file/d/1tboZOkfaIK6kNStfGcxVMAX5FegiDrQ9/view?usp=drive_link

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Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

- Python 3.10 or above
- Git
- Flask
- Required Python libraries (listed in `requirements.txt`)
- Visual Studio Code (recommended)

Steps to Run the Project

1. Clone the GitHub repository.
2. Open the project in Visual Studio Code.
3. Create and activate the virtual environment.

.\venv\Scripts\Activate.ps1

4. Install the required dependencies:

Pip install –r requirements.txt

5. Start the Flask application:

Python app.py

6. Open the following URL in the browser

http://127.0.0.1:5000

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Application is currently deployed only on the local system.	Can be deployed on Render, AWS, or Azure in the future.
2	Prediction accuracy depends on the quality of the input data.	Users should provide accurate soil and weather values.
3	Weather API and Fertilizer Recommendation modules are not yet integrated.	Planned as future enhancements.